

Project Development Phase

The **Project Development Phase** is an important stage in the creation of **EduGenie – Google Gemini Powered Learning Assistant**. In this phase, the actual coding, integration, and implementation of the system are carried out based on the project design. The aim of this phase is to transform the planned ideas into a working application.

1. Development Objectives

The main objectives of the development phase are:

- To build the user interface of the application.
- To implement the backend functionalities.
- To integrate Google Gemini API for AI-based responses.
- To develop modules for doubt solving and user interaction.
- To ensure smooth communication between all components.

2. Frontend Development

The frontend is responsible for interacting with the users. It provides a simple and user-friendly interface where students can ask questions and receive answers.

The frontend may include:

- Home page
- Login/Register page
- Chat interface
- Question input area
- Response display section
- Conversation history page

Technologies used:

- HTML
- CSS
- JavaScript

3. Backend Development

The backend handles the processing of user requests and communication with the AI model.

Backend functions include:

- Receiving user queries
- Sending requests to Google Gemini API
- Processing responses
- Managing user information
- Storing conversation history

Technologies used:

- Python
- Flask/Django (if required)

4. AI Integration

Google Gemini API is integrated into the application to provide intelligent responses.

The integration process includes:

1. User enters a question.
2. The system sends the question to Gemini API.
3. Gemini processes the query.
4. AI generates an appropriate answer.
5. The response is displayed to the user.

5. Database Development

A database can be used to store:

- User details
- Questions asked
- AI-generated responses
- Date and time information
- Chat history

Possible databases:

- MySQL
- Firebase
- SQLite

6. Module Implementation

The project is developed using different modules:

- **User Module:** Handles login and registration.
- **Doubt Solving Module:** Processes student questions.
- **AI Module:** Communicates with Google Gemini.

- **Database Module:** Stores information.
- **Response Module:** Displays answers.

7. Testing During Development

Regular testing is performed during development to ensure:

- Correct functionality
- Fast response time
- Accurate outputs
- Error-free operation
- User-friendly experience

8. Expected Outcome

The Project Development Phase converts the project design into a fully functional application. This phase ensures that **EduGenie works efficiently as an AI-powered learning assistant**, helping students solve doubts quickly and effectively.